

Leakage Current Analysis of GaN-Based Light-Emitting Diodes Using a Parasitic Diode Model

Eunjin Jung, June Key Lee, Min Soo Kim, and Hyunsoo Kim

Abstract—The leakage current of GaN-based light-emitting diodes (LEDs) was analyzed with a parasitic diode model. Arrhenius plots of forward leakage currents revealed that the threshold voltages of the main diode and the parasitic diode were 2.64 and 0.94 V, respectively. The parasitic diode, however, led to an insignificant change in the reverse leakage mechanism, e.g., a predominant reverse leakage mechanism was Poole–Frenkel conduction in the high-temperature range (>300 K) and the variable range hopping in the low-temperature range (<300 K), as reported previously. The origin of the parasitic diode could be due to the incorporated hydrogen (H) atoms into the top LEDs layers, presumably enhancing the defect-assisted tunneling through the generated H-related deep-level states, particularly generated in the p-cladding or p-AlGaIn electron blocking layer.

Index Terms—Hydrogen (H)-related deep-level states, leakage current, light-emitting diode (LED), parasitic diode.

I. INTRODUCTION

AN EXACT understanding of the leakage current mechanism in GaN-based light-emitting diodes (LEDs) is of particular importance due to its impact on the reliability of a device [1]. In general, the parasitic shunt resistance bypassing the main diode was responsible for the evolution of leakage currents [2]–[6]. The origin of the parasitic shunt resistance acting as a parallel resistor was suggested to be associated with the diffused ions [7] and/or the structural imperfections [8], such as threading dislocations and point defects in the depletion region. Consequently, the leakage mechanism could be well understood in terms of hopping conduction in the

low-temperature range and the field-assisted thermal ionization or the so-called Poole–Frenkel emission in the high-temperature range [5], [6].

Unlike this prevalent leakage mechanism, Kim *et al.* [9] also showed that the exposure of LED wafers in a hydrogen (H) ambient could produce distinctive leakage behavior, which is solely observed in the forward bias region (looks like a hump), not in the reverse bias, namely, the formation of a parasitic diode. As a possible origin of the generated parasitic diode, the incorporation of H atoms into the p-GaN, leading to a formation of H-related deep-level states in the p-cladding layer, was suggested. However, this leakage mechanism should be further and more clearly understood, since the conventional LED process employs H gases, e.g., for the deposition of SiO₂ film by plasma-enhanced chemical vapor deposition (PECVD). It will also be meaningful to investigate whether the conventional leakage mechanisms, such as hopping and the Poole–Frenkel model, are influenced by the presence of parasitic diodes. In addition, it should be also addressed whether the H reduction in the p-cladding layer would suppress the generation of parasitic diodes, i.e., whether the forward hump is truly due to the H-related deep-level states in p-GaN. However, in-depth studies on this subject are very rare. In this regard, we investigated the leakage current mechanism of GaN-based LEDs with generated parasitic diodes.

II. EXPERIMENTAL PROCEDURES

Commercially available GaN/InGaIn LED wafers with a 385-nm emission wavelength were used to investigate the electrical properties. For the fabrication of LEDs, SiO₂ films were deposited on the top p-GaN surfaces of the LED wafers using a PECVD, followed by the formation of a $435 \times 270\text{-}\mu\text{m}^2$ mesa by subsequent photolithography and dry etching using an inductively coupled plasma etching system. Note that during SiO₂ deposition, the top p-GaN layer was exposed to H gas, which was decomposed from a SiH₄ source. A conventional Indium Tin Oxide (ITO) contact was formed on the p-GaN as a transparent electrode, followed by a rapid thermal annealing for ohmic contact. A Cr/Au contact was last formed on the n-layer and the p-GaN/ITO layer as the n-electrode and the p-bonding pads, respectively. All contacts were deposited using an e-beam evaporator. The schematic cross-sectional view of the lateral-injection LEDs are shown in Fig. 1. For reliable measurements, five LEDs were packaged using the 9280 surface mount device mold. In addition, the LED wafers treated with electrochemical

Manuscript received January 23, 2015; revised July 30, 2015; accepted August 11, 2015. Date of publication August 25, 2015; date of current version September 18, 2015. This work was supported in part by the Ministry of Trade, Industry and Energy through the Transfer Machine Specialized Lighting Core Technology Development Professional Manpower Training Project under Grant N0001363, in part by the Basic Science Research Program within the National Research Foundation (NRF) through the Ministry of Education, Science and Technology under Grant NRF-2014R1A1A1A05007455, Korea, and in part by the Ministry of Education and NRF through the Human Resource Training Project for Regional Innovation under Grant 2013H1B8A2032197. The review of this paper was arranged by Editor K. J. Chen. (Corresponding authors: June Key Lee, Min Soo Kim, and Hyunsoo Kim.)

E. Jung and H. Kim are with the Semiconductor Physics Research Center, School of Semiconductor and Chemical Engineering, Chonbuk National University, Jeonju 61-756, Korea (e-mail: jerrysco@jbnu.ac.kr; hskim7@jbnu.ac.kr).

J. K. Lee is with the Optoelectronics Convergence Research Center, Department of Materials Science and Engineering, Chonnam National University, Gwangju 500-757, Korea (e-mail: junekey@jnu.ac.kr).

M. S. Kim is with the Department of IT Convergence Engineering, Daejeon University, Daejeon 300-716, Korea (e-mail: minsoo@dju.ac.kr).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TED.2015.2468581

0018-9383 © 2015 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.

See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

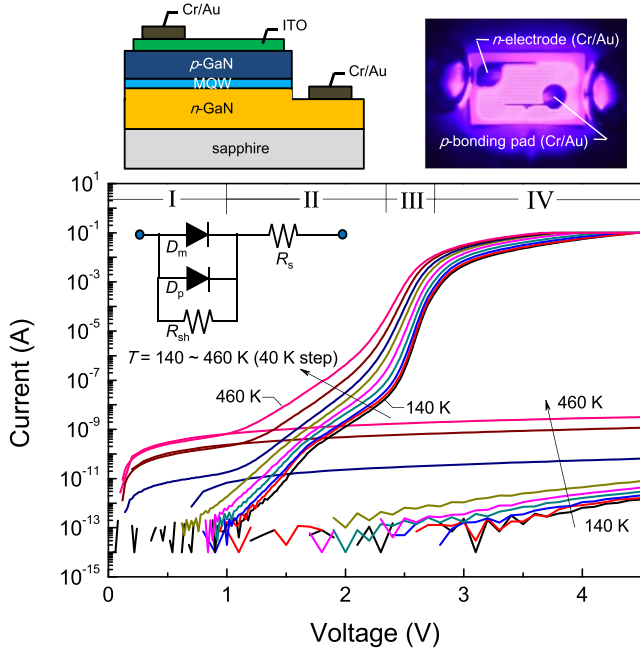


Fig. 1. I - V - T characteristics of LEDs. Insets: schematic LED structure, top electroluminescent images, and equivalent electric circuit.

potentiostatic activation (EPA) were also fabricated to investigate the effect of H on the electrical properties [10], [11]. A secondary ion mass spectroscopy (SIMS) depth profiling was performed before and after EPA treatment to investigate the change in the atomic H concentration [H], which was estimated to be 8.1×10^{19} and $3.3 \times 10^{19} \text{ cm}^{-3}$ for the p-GaN layer before and after EPA treatment, respectively. The fabricated LEDs were evaluated using a parameter analyzer (HP4156A) and a temperature variable probe station (140–460 K).

III. RESULTS AND DISCUSSION

Fig. 1 shows the typical current–voltage–temperature (I - V - T) characteristics of LEDs. Evidently, four distinctive domains were observed in the forward I - V curves: 1) I ($V < 1$ V); 2) II ($1 \text{ V} < V < 2.34$ V); 3) III ($2.34 \text{ V} < V < 2.76$ V); and 4) IV ($V > 2.76$ V) for $T = 300$ K. This is well explained in terms of a parasitic diode circuit model as shown in Fig. 1 (inset), where the circuit consists of the main diode (D_m : predominant in domain III), parasitic diode (D_p : domain II), shunt resistance (R_{sh} : domain I), and series resistance (R_s : domain IV) [3], [9]. As mentioned above, the appearance of the D_p in domain II seems to be associated with the incorporated H atoms, as will be discussed later.

To investigate the transport mechanism in domain I, the reverse leakage currents were plotted as a function of the reciprocal temperature (since R_{sh} has no bias-polarity dependence), as shown in Fig. 2. Note that the Arrhenius plot was available in the high-temperature range ($T \geq 340$ K), yielding a thermal activation energy (E_A) of 729 meV according to [5]

$$I \propto \exp\left(-\frac{E_A}{kT}\right) \quad (1)$$

where k is the Boltzmann constant. The obtained E_A is much smaller than the half bandgap energy ($E_g/2$),

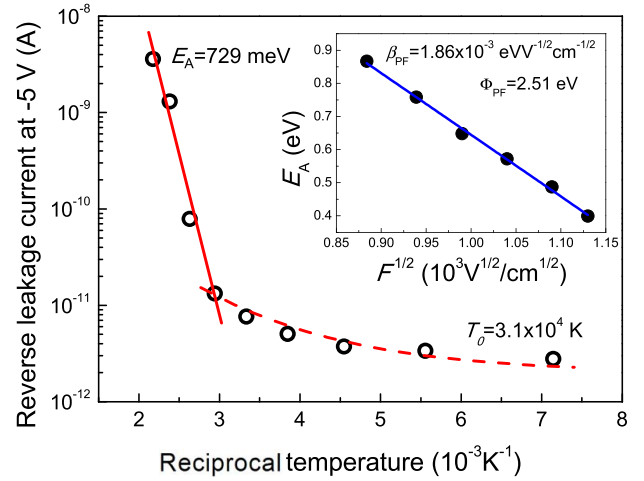


Fig. 2. Arrhenius plot of reverse leakage current at -5 V. Inset: E_A as a function of the square root of the average internal electric field.

indicating that the classical drift-diffusion and/or generation–recombination model cannot explain carrier transport in domain I [3]. Indeed, E_A showed a strong field dependence, as shown in Fig. 2 (inset), indicating Poole–Frenkel conduction [12], [13]

$$E_A = \Phi_{PF} - \beta_{PF} F^{1/2} \quad (2)$$

where Φ_{PF} is the barrier height of carriers trapped in deep-level states without an external electric field, β_{PF} is the Poole–Frenkel constant, and F is the local electric field strength exerted on the deep-level states. Here, F is defined as [5]

$$F = \frac{(V_{BI} - V)}{W_d} \quad (3)$$

where V_{BI} is the built-in voltage of LED (~ 3.4 V) [5] and W_d is the depletion width at zero bias. The W_d was obtained to be 101 nm based on the capacitance–voltage analysis (not shown). Note that, as shown in Fig. 2 (inset), the linear fit yielded that $\Phi_{PF} = 2.51$ eV and $\beta_{PF} = 1.86 \times 10^{-3} \text{ eV V}^{-1/2} \text{ cm}^{1/2}$, which are in the typical ranges as reported previously [2], [4], [6]. This indicates that the leakage mechanism in domain I was not influenced by the presence of the D_p . This argument was further supported by the consistent leakage mechanism in the low-temperature range ($T < 300$ K), in which the variable range hopping conduction dominated, since the reverse leakage current followed $T^{-1/4}$ dependence [14]

$$I \propto \exp\left\{\left(-\frac{T_0}{T}\right)^{1/4}\right\} \quad (4)$$

where T_0 is the characteristic temperature (3.1×10^4 K). Therefore, it is concluded that one of the implicit assumptions of the D_p model, i.e., the forward hump does not affect reverse leakage mechanism, is applicable [9].

To investigate the carrier transport in domains II and III, the forward currents (at a certain bias voltage) were also plotted as a function of reciprocal temperature, as shown in Fig. 3. For example, at 1.5 V, Arrhenius plots of the forward currents

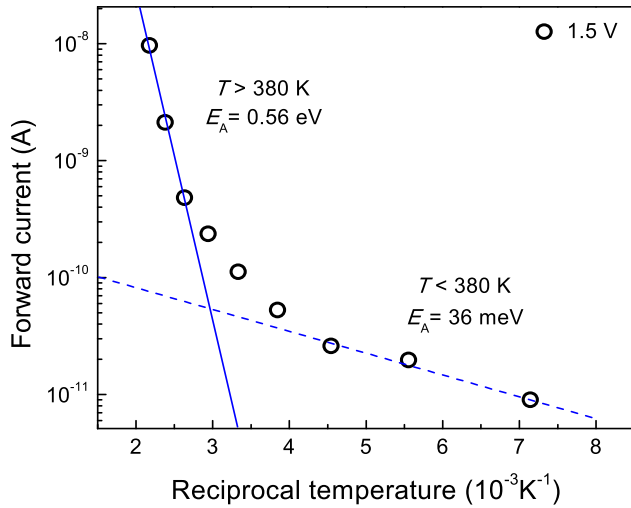


Fig. 3. Typical Arrhenius plot of forward current measured at 1.5 V.

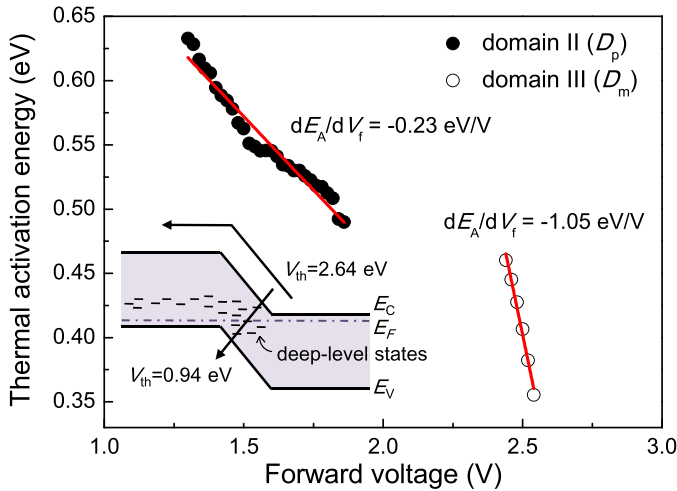


Fig. 4. E_A versus V_f plots. Inset: schematic band diagram depicting the carrier transport in domains II and III.

were available in the high-temperature ($T > 380$ K) and low-temperature ranges ($T < 380$ K), given that $E_A = 0.56$ eV and 36 meV, respectively (see Fig. 3). Note that the E_A obtained in the low-temperature ranges was negligibly small, indicating that tunneling is predominant [3]. Therefore, the E_A value obtained in the high-temperature range is expected to reflect the actual barrier height of the diodes.

Fig. 4 summarizes the E_A values (obtained in the high-temperature range) as a function of forward bias voltage (V_f). It is evident that the E_A has a significant negative bias dependence [3], e.g., $dE_A/dV_f = -0.23$ and -0.89 eV/V for domain II (D_p) and domain III (D_m), respectively. According to [3], using the voltage dependence of the E_A values (dE_A/dV_f), the energy barrier for which the carriers to overcome or the threshold voltage (V_{th}) could be obtained by the extrapolation at zero bias. Using the same method, the extrapolation of our experimental data yielded that $V_{th} = 0.94$ eV for D_p (domain II) and $V_{th} = 2.64$ eV for D_m (domain III).

In principle, V_{th} of D_m should be equivalent to the built-in potential of the p-n junction (3.4 eV), since the diffusion

current dominates in domain III. Accordingly, the obtained V_{th} of D_m (2.64 eV) was somewhat in good agreement with the built-in potential. According to the empirical diode equation [15], i.e., $I = I_0[\exp(qV/nkT) - 1]$, where I_0 is the saturation current and n is the ideality factor, the obtained n value in domain III was as low as 1.41. This n value, which is close to 1.0, indicates that the predominant transport mechanism in domain III is diffusion. This is consistent with the obtained large barrier height of D_m , since the carrier should overcome the junction voltage via diffusion, as described in Fig. 4 (inset).

For the D_p , however, V_{th} of 0.94 eV was much smaller than that of D_m , indicating that there is easier pathway for carrier transport associated with the deep-level states [see Fig. 4 (inset)]. Indeed, a similar forward hump leakage was observed in [9], reporting that the enhanced carrier transport might be due to the so-called defect-assisted tunneling. In particular, Mandurrino *et al.* [16] classified the defect-assisted tunneling into three processes, i.e., defect-assisted tunneling through the quantum wells (QWs), defect-to-defect tunneling under the QWs, and defect-assisted tunneling between QWs barrier and p-type electron blocking layer (EBL). Huang *et al.* [17] showed that the multistep tunneling can occur through the localized deep-level states residing in EBL and p-GaN cladding layer, which were produced during the dry etching and thermal bonding processes. As a possible origin of deep-level states, Ga vacancies, N vacancies, Mg vacancy complexes ($Mg_{Ga}-V_N$), oxygen (O) ions, carbon, and other related complexes were suggested. Similarly, Cao *et al.* [15] also proposed that the O impurities, N vacancies, Mg-H complex, and structural defects in the barrier/well interfaces of QWs and the interface of QW/cladding layer might be responsible for the defect-assisted tunneling. Meanwhile, Yan *et al.* [18] suggested that the defect-assisted tunneling is a limited process occurring at space charge region, in which electron could move diagonally from conduction band of n-side to valence band of p-side in E_g of multiple QWs. These previous studies suggest that the deep-level states acting as the pathway for carrier transport with lower V_{th} can reside in the QWs active region, EBL, and p-cladding layer.

On the one hand, according to [9], the generation of D_p was caused by the incorporated H atoms. To find the effect of H atoms on the forward hump leakage, the EPA process was performed, as shown in Fig. 5. Fig. 5 also shows the forward I - V curves of the reference LEDs and EPA-treated LEDs. The fundamental concept of EPA treatment is breaking the Mg-H bonds and taking out H atoms from the inside of LED structures, particularly from the p-GaN of p-AlGaN EBL, by applying an electric field under an electrolyte solution [10], [18]. Note that the EPA treatment was very useful for extracting H atoms in the p-GaN layer, as shown in Fig. 5 (inset), where the H was reduced by 59% after EPA treatment. Here, the SIMS depth profiling was performed for the p-GaN layer (prepared for the SIMS analysis, not LEDs) before and after EPA treatment. Consistently, the forward hump was significantly reduced. Therefore, it is concluded that the H-related deep-level state in p-GaN cladding layer or p-AlGaN EBL layer might be associated with the generation of D_p .

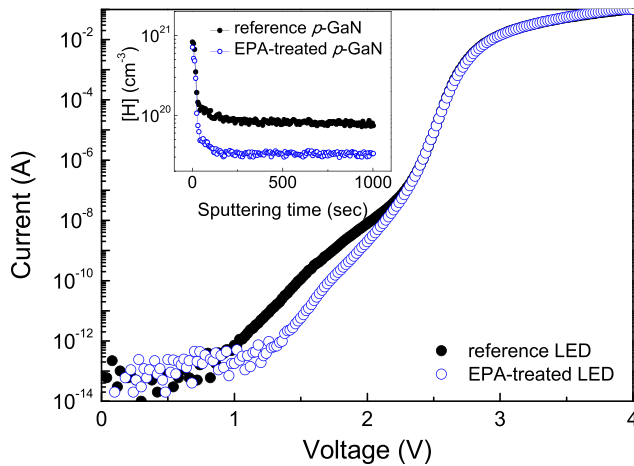


Fig. 5. Forward I - V curves of reference LEDs and EPA-treated LEDs. EPA treatment was performed at 7 V for 5 min in 1.0 M HCl solution. Inset: H concentration $[H]$ before and after EPA treatment, as obtained from the SIMS depth profiling.

Meanwhile, as reported in [11], the EPA treatment of LED structure might generate another type of point defects, such as N vacancies in the active region by breaking the Ga-H bonds and subsequently extracting H atoms (incorporated into the active region during epitaxial growth). In this case, the EPA process would give a negative influence on the forward hump leakage (will increase the hump) since the generated N vacancies can act as an additional deep-level states (enhancing the defect-assisted tunneling). Therefore, it is argued that D_p is more closely related to the presence of H-related deep-level states in the p-GaN cladding and/or p-AlGaIn EBL layers than the deep-level states in the active region. However, a more precise mechanism is presently under investigation.

Finally, it would be meaningful to investigate the relation between the optical output power (L) and the tunneling current region associated with D_p (10^{-7} to 10^{-9} A). But, we failed to observe the L data at such a tunneling current range presumably due to the poor defection limit of the optical detector, and more predominantly, due to the nonradiative recombination dynamics as the tunneling process dominates.

IV. CONCLUSION

The carrier transport of GaN-based LEDs was analyzed in terms of a parasitic diode model. The threshold voltage of the parasitic diode (0.94 eV) was much lower than that of the main diode (2.64 eV), which was attributed to the carrier transport through deep-level states of the p-cladding layer. The presence of the parasitic diode, however, caused insignificant change in the reverse leakage mechanism. The generation of the parasitic diode was due to H incorporation into the p-cladding layer, as verified by the EPA treatment and the SIMS analysis.

REFERENCES

- [1] M. Meneghini, A. Tazzoli, G. Mura, G. Meneghesso, and E. Zanoni, "A review on the physical mechanisms that limit the reliability of GaN-based LEDs," *IEEE Trans. Electron Devices*, vol. 57, no. 1, pp. 108–118, Jan. 2010.
- [2] D. V. Kuksenkov, H. Temkin, A. Osinsky, R. Gaska, and M. A. Khan, "Origin of conductivity and low-frequency noise in reverse-biased GaN p-n junction," *Appl. Phys. Lett.*, vol. 72, no. 11, pp. 1365–1367, Mar. 1998.

- [3] L. Hirsch and A. S. Barrière, "Electrical characterization of InGaIn/GaN light emitting diodes grown by molecular beam epitaxy," *J. Appl. Phys.*, vol. 94, no. 8, pp. 5014–5019, Oct. 2003.
- [4] M. S. Ferdous, X. Wang, M. N. Fairchild, and S. D. Hersee, "Effect of threading defects on InGaIn/GaN multiple quantum well light emitting diodes," *Appl. Phys. Lett.*, vol. 91, no. 23, pp. 231107-1–231107-3, Dec. 2007.
- [5] Q. Shan *et al.*, "Transport-mechanism analysis of the reverse leakage current in GaInN light-emitting diodes," *Appl. Phys. Lett.*, vol. 99, no. 25, pp. 253506-1–253506-3, Dec. 2011.
- [6] J. Kim *et al.*, "Investigation of reverse leakage characteristics of InGaIn/GaN light-emitting diodes on silicon," *IEEE Electron Device Lett.*, vol. 33, no. 12, pp. 1741–1743, Dec. 2012.
- [7] A. Mao, J. Cho, Q. Dai, E. F. Schubert, J. K. Son, and Y. Park, "Characteristics of dotlike green satellite emission in GaInN light emitting diodes," *Appl. Phys. Lett.*, vol. 98, no. 2, pp. 023503-1–023503-3, Jan. 2011.
- [8] L. Lymperakis, J. Neugebauer, M. Albrecht, T. Remmele, and H. P. Strunk, "Strain induced deep electronic states around threading dislocations in GaN," *Phys. Rev. Lett.*, vol. 93, no. 19, pp. 196401-1–196401-4, Nov. 2004.
- [9] H. Kim, J. Cho, Y. Park, and T.-Y. Seong, "Leakage current origins and passivation effect of GaN-based light emitting diodes fabricated with Ag p-contacts," *Appl. Phys. Lett.*, vol. 92, no. 9, pp. 092115-1–092115-3, Mar. 2008.
- [10] M. Oh, J. J. Lee, J. K. Lee, and H. Kim, "Electrical characteristics of Mg-doped p-GaN treated with the electrochemical potentiostatic activation method," *J. Alloy Compounds*, vol. 585, pp. 414–417, Feb. 2014.
- [11] J. K. Lee *et al.*, "Electrochemical removal of hydrogen atoms in Mg-doped GaN epitaxial layers," *J. Appl. Phys.*, vol. 117, no. 18, pp. 185702-1–185702-7, May 2015.
- [12] J. Frenkel, "On pre-breakdown phenomena in insulators and electronic semi-conductors," *Phys. Rev.*, vol. 54, pp. 647–648, Oct. 1938.
- [13] J. Kim, J. Kim, Y. Tak, S. Chae, J.-Y. Kim, and Y. Park, "Effect of V-shaped pit size on the reverse leakage current of InGaIn/GaN light-emitting diodes," *IEEE Electron Device Lett.*, vol. 34, no. 11, pp. 1409–1411, Nov. 2013.
- [14] N. F. Mott, "Conduction in non-crystalline materials," *Philos. Mag.*, vol. 19, no. 160, pp. 835–852, Apr. 1969.
- [15] X. A. Cao, E. B. Stokes, P. M. Sandvik, S. F. LeBoeuf, J. Kretschmer, and D. Walker, "Diffusion and tunneling currents in GaN/InGaIn multiple quantum well light-emitting diodes," *IEEE Electron Device Lett.*, vol. 23, no. 9, pp. 535–537, Sep. 2002.
- [16] M. Mandurriño *et al.*, "Trap-assisted tunneling in InGaIn/GaN LEDs: Experiments and physics-based simulation," in *Proc. 14th Int. Conf. NUSOD*, Palma de Mallorca, Spain, Sep. 2014, pp. 13–14.
- [17] Y. Huang *et al.*, "Defect-induced color-tunable monolithic GaN-based light-emitting diodes," *Appl. Phys. Exp.*, vol. 7, no. 10, pp. 102102-1–102102-4, Sep. 2014.
- [18] D. Yan, H. Lu, D. Chen, R. Zhang, and Y. Zheng, "Forward tunneling current in GaN-based blue light-emitting diodes," *Appl. Phys. Lett.*, vol. 96, no. 8, pp. 083504-1–083504-3, Feb. 2010.

Eunjin Jung received the M.S. degree from Chonbuk National University, Jeonju, Korea, where she is currently pursuing the Ph.D. degree with the School of Semiconductor and Chemical Engineering.

June Key Lee received the Ph.D. degree from the Virginia Polytechnic Institute and State University, Blacksburg, VA, USA.

He has been with the Department of Materials Science and Technology, Chonnam National University, Gwangju, Korea, since 2004.

Min Soo Kim received the Ph.D. degree from the Korea Advanced Institute of Science and Technology, Daejeon, Korea.

He has been with the Department of IT Convergence Engineering, Daejeon University, Daejeon, since 2015.

Hyunsoo Kim received the Ph.D. degree from the Gwangju Institute of Science and Technology, Gwangju, Korea.

He has been with the Department of Semiconductor Science and Technology, Chonbuk National University, Jeonju, Korea, since 2009.